

Secondary Science

User Guide

Welcome to Twinkl Secondary Science!

- Twinkl Secondary Science gives you **24/7 access** to an ever-expanding collection of **essential planning** and **teaching resources**.
- Discover thousands of materials, with new KS3 and KS4 resources added daily – all tailored to closely match the **National Curriculum** and **GCSE specifications**.
- All resources are **expertly created** by Twinkl's team of **experienced** and hands-on **secondary teachers**.
- **Whatever your challenge** and **whatever your need**, find the perfect planning, teaching or learning resources right here.

To help you make the most of your Twinkl Secondary Science subscription, this handy guide introduces you to the huge range and scope of resources now at your fingertips. Browse the pages and discover how we can help you deliver a full and flexible curriculum.

We've tried to answer any questions you might have. However, if you find there's something we haven't covered, simply **get in touch** with your friendly Twinkl Cares team – we're happy to help.

Love Twinkl x

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Secondary Solutions at Your Fingertips

We help those who teach.

Twinkl's mission is to 'help those who teach' and although we're perhaps best known for the success of our primary resources, we're every bit as dedicated to supporting the incredible work of secondary teachers.

That's why we employ our own team of experienced secondary teachers to create a broad range of high-quality materials for the KS3/4 curriculum.

- We're proactive, tracking government curriculum changes to build a library of topics, lesson packs and exam materials that meet all assessment objective requirements.
- New resources are added each day, providing a flexible and adaptable toolkit of secondary solutions for the ongoing demands of planning, teaching and marking.

With such high-quality versatile content at your fingertips, you'll have more time to really focus on your classroom delivery.

We're constantly working to expand the variety and breadth of our units. As always, we follow National Curriculum and exam board aims and objectives, so you'll find that our work increasingly refers to these.

That's not all! You'll also find a growing collection of high-quality digital media resources, giving you even more powerful and engaging ways to inspire your pupils. This constant innovation is what makes Twinkl Secondary Science such an incredible teaching tool.



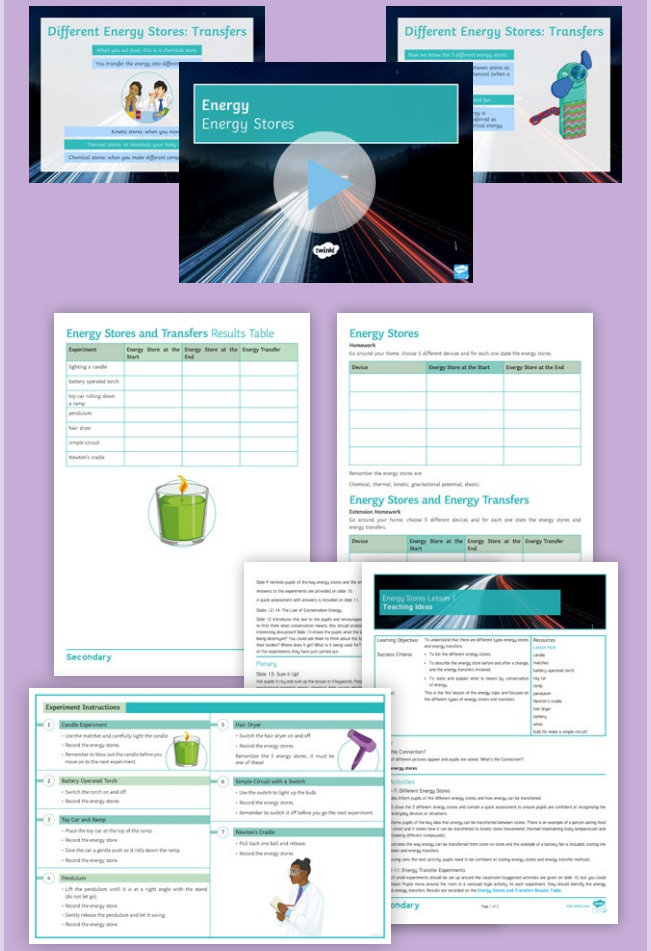
Your Secondary Teaching Partner

Enhance your pedagogy

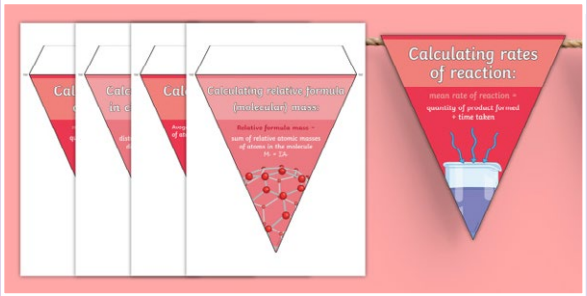
With this ever-expanding suite of new ideas and concepts available 24/7, you can seamlessly blend your own teaching practice with ours, using our teacher-made resources to bring variety and invention to your classroom.

- Whenever you require content that complements your own strategies, you'll find it right here.
- Classroom quality guaranteed! All resources created by experienced and active Secondary teachers.
- All resources match National Curriculum and exam board aims.
- Every Twinkl Secondary Science resource is available digitally and can be downloaded at any time, 24/7.

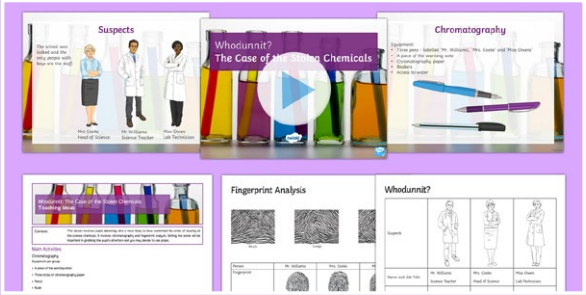
Energy Changes and Transfers Lesson 1: Energy Stores



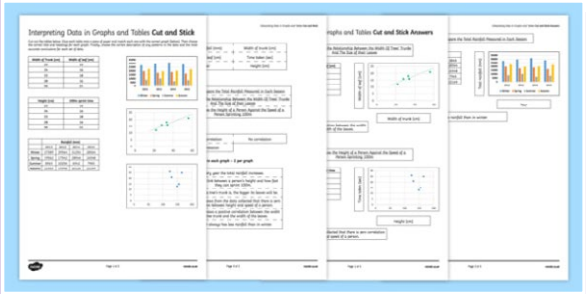
GCSE Chemistry - Written Equations Display Bunting



Whodunnit? Scientific Investigation Lesson Pack



Interpreting Data in Graphs and Tables Activity Sheet



Units of Work

Detailed lessons, learning objectives and more

Twinkl Secondary units of work are intended to be versatile. As you'll see, they're numbered sequentially following the GCSE Science courses closely. Of course, you're not bound by this order, so feel free to swap and change; dipping in and out in ways that suit you. You're free to use all Twinkl resources however you like.

To help you even more, a number of these units of work come with an associated unit overview. This details all the lessons, plus their learning objectives and/or their relationship to the National Curriculum.

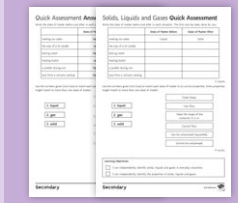
Each unit of work also contains a number of teacher-made lesson packs, saving you time and any additional workload.

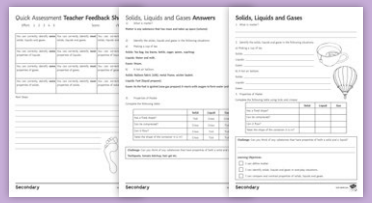
- Lesson packs are **thoroughly researched** and **intended to be exhaustive** -we deliberately include more material than you'll need for one lesson.
- Pick and choose the ideas that **best suit your class**, plus **tailor your lesson** according to the pupils you're working with - giving you the **flexibility** to roll learning into the next lesson.
- Lessons provide practical ideas and suggested activities to **engage learners** and **help with challenging content**.

Our general lessons have been created by experienced secondary teachers, using high-quality content that's designed to be used by subject experts like you.

States of Matter Lesson 1: Solids, Liquids and Gases





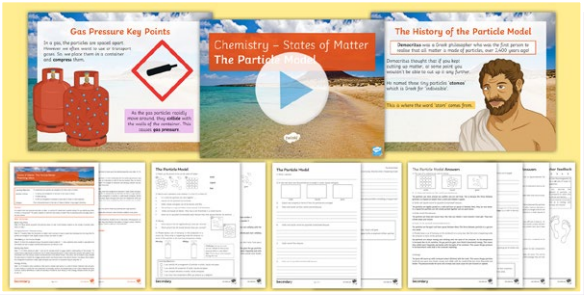




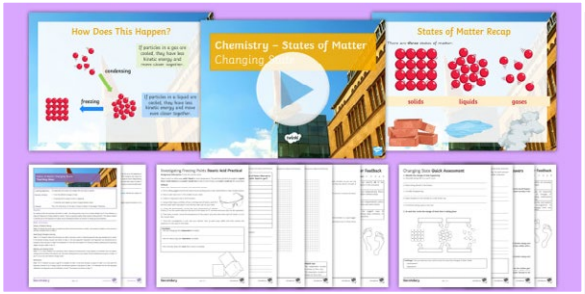
What's more, Secondary Science lesson packs are so detailed that you can also use them to set for cover - simply print off the teacher notes then highlight the areas that you want to be taught by your cover teacher.

- Once you've accessed a lesson pack page, scroll down to the "linked to this resource" box.
- You'll find direct links back to the unit overview, and to the subsequent and antecedent lessons in the unit sequence.
- Individual lesson packs include a PowerPoint, teacher notes and at least one activity sheet option, plus opportunities for differentiation.

States of Matter Lesson 2: The Particle Model



States of Matter Lesson 3: Changing States

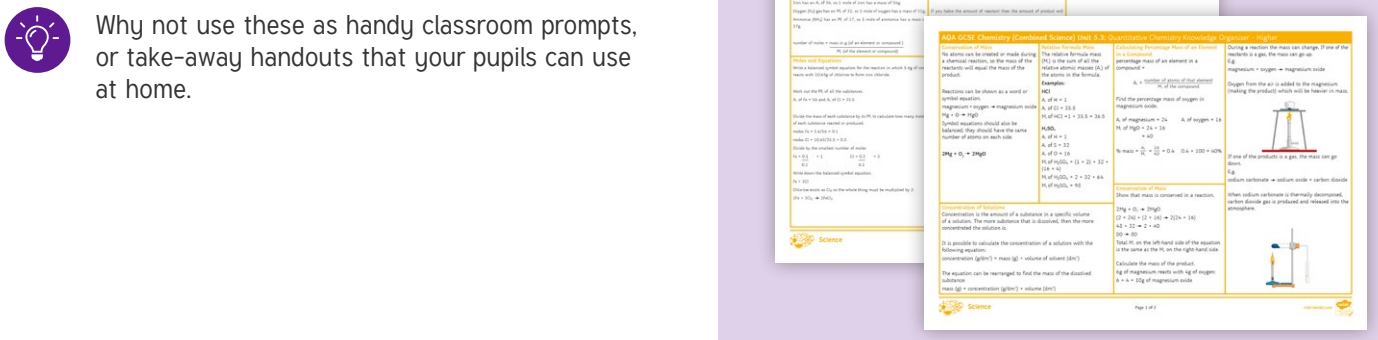


Knowledge Organisers

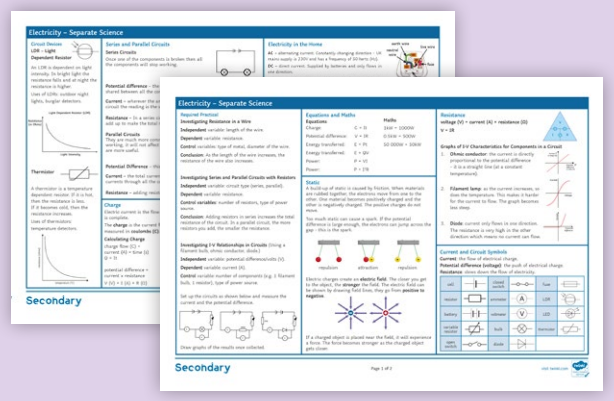
Versatile teaching and learning tools

Another useful tool is the knowledge organiser. This versatile resource will be available for every major topic and contains all the key pieces of information in bite-sized chunks, across one double-sided A4 sheet.

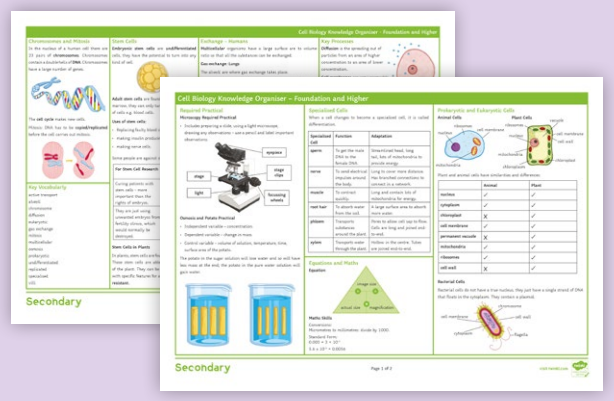
Why not use these as handy classroom prompts, or take-away handouts that your pupils can use at home.



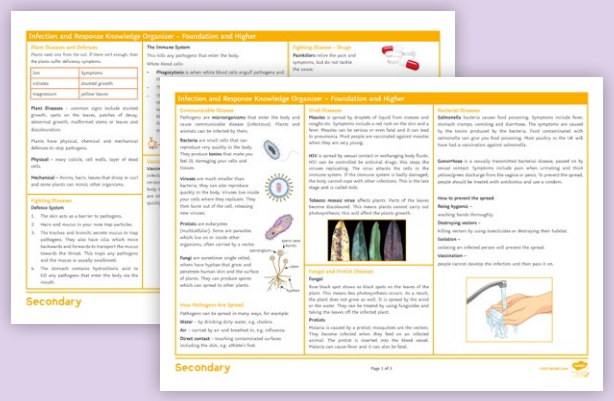
Knowledge Organiser: Quantitative Chemistry



Knowledge Organiser: Electricity



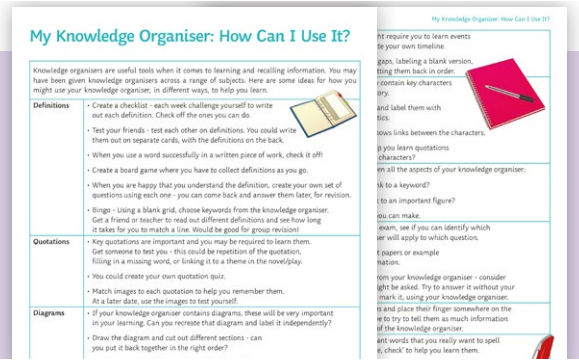
Knowledge Organiser: Cell Biology



Knowledge Organiser: Infection and Response

Explore the following resource for other great ways to use Twinkl's knowledge organisers - perfect for classroom prompts and an easy-to-use revision tool!

My Knowledge Organiser: How Can I Use It?



Assessment Aids

Developing skills for test environments

Revision Mats

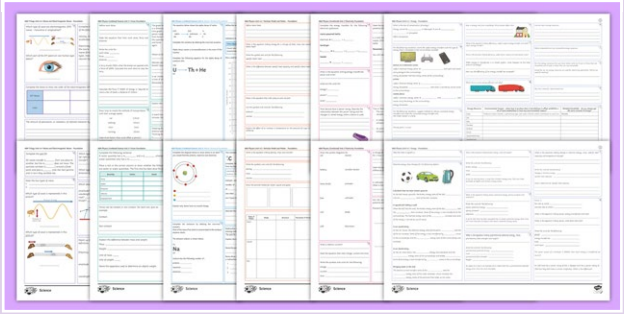
Twinkl Secondary revision mats are a fantastic resource for end of unit exams, mocks and GCSE exams. These are specific to each unit of work and follow the curriculum and exam board specifications.

With sets of revision questions on just two sides of A4, you'll find they're simple to use and easy to print. Questions range from short to long answer, and include the required practical exercises too.

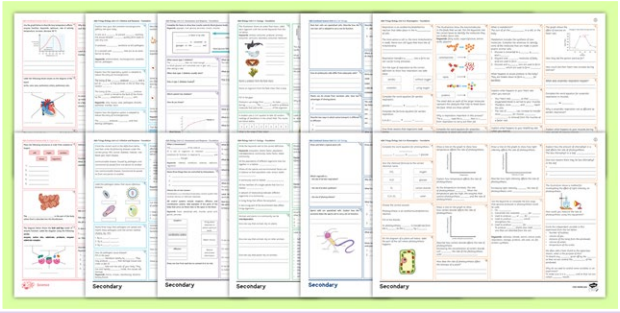
AQA Chemistry Foundation Revision Mat Bundle



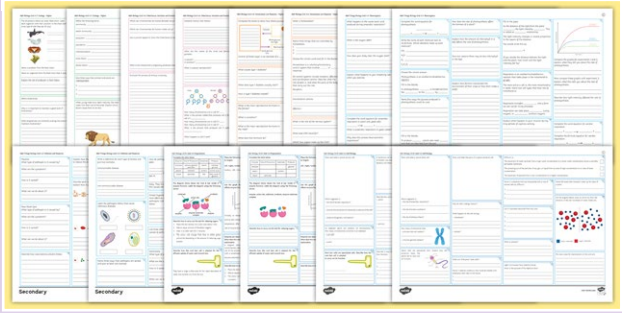
AQA Physics Foundation Revision Mat Bundle



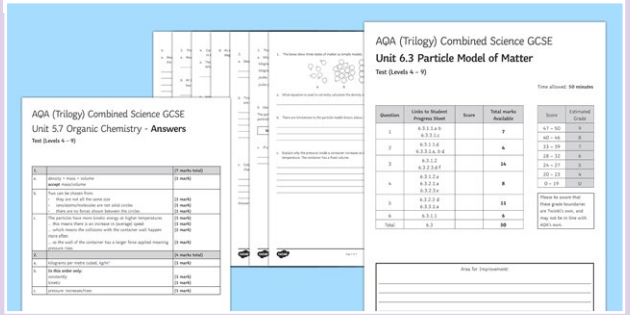
AQA Biology Foundation Revision Mat Bundle



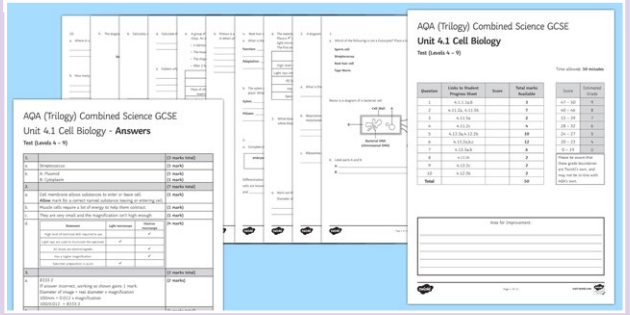
AQA Biology Higher Revision Mat Bundle



AQA Style Paper: Unit 3 Particle Model of Matter



AQA Style Paper: Unit 1 Cell Biology



End of Unit Tests

Each unit of work includes end of unit tests to help with exam practice. As you'd expect, the questions have been thoroughly researched to make sure they're fully in line with the GCSE exam boards. Plus, they include application and mathematical questions too, just as GCSE exam boards do.

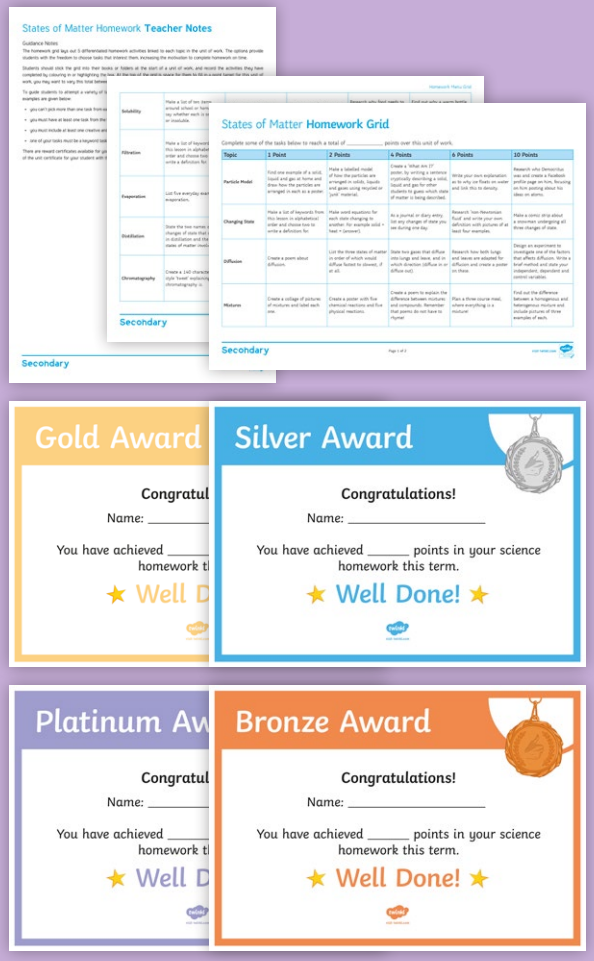
Homework Grids

Extend learning beyond the classroom

You'll also find homework grids accompanying each unit of work. These grids are linked to each lesson within the unit of work and pupils can choose from a range of questions or activities to gain points as they work through the unit – questions carry scores of 1, 2, 3, 6 or 10 points.

For added engagement, we also include two separate reward systems for both KS3 and GCSE pupils, plus certificates for KS3 pupils and positive postcards for GCSE. Rewards can be given out once the unit has been completed.

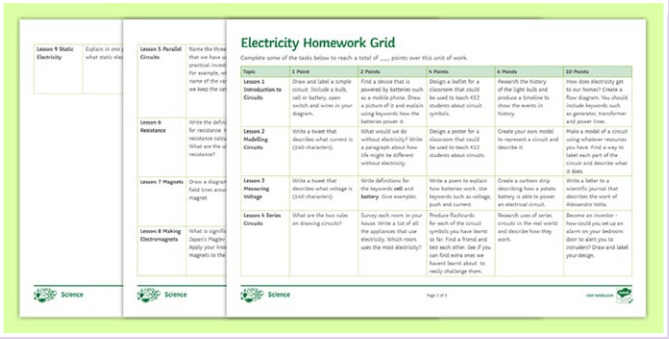
States of Matter Homework Grid



Cells and Organisation Homework Grid



Electricity Homework Grid



Homework grids come with answers for ease of marking and you can choose how many points pupils are to achieve through the unit - perfect for setting differentiated homework.

Specimen Exam Material

Helping you prepare your pupils

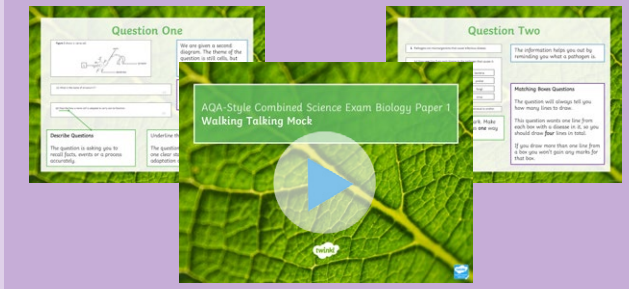
It's not unusual for KS4 Science teachers to run out of specimen exam papers by the mid-way point of the year. So to help, Twinkl Secondary Science gives you access to specimen papers in the style of each exam board, including AQA, Edexcel and OCR. Questions are worded to reflect each boards' specifications and each pack includes a question paper and mark scheme.

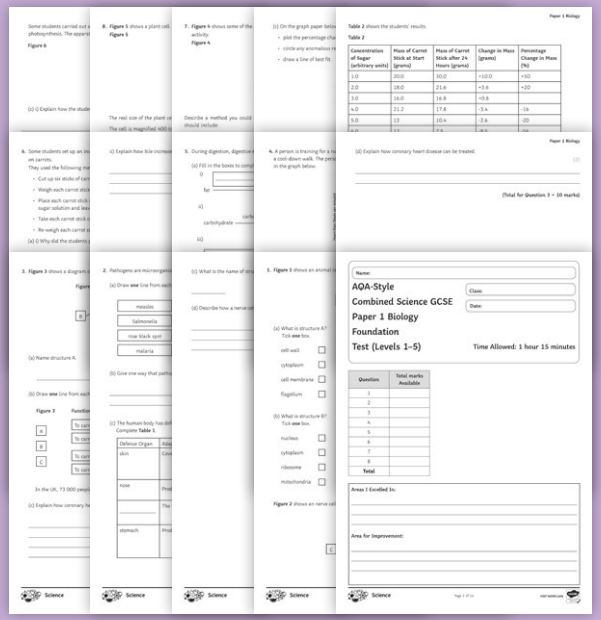
Our "Walking Talking Mocks" help you lead your classes through science exam questions, under timed conditions. These packs include an exam paper and a PowerPoint, providing a fantastic way to prepare pupils for sitting exams, with teacher-led discussions between each question.

Use our specialist exam resources to acclimatise your pupils with a taste of real exam challenges and experiences.

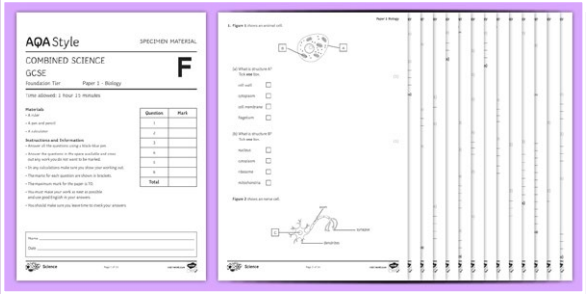
Current GCSE specifications lean heavily on Assessment Objectives, so these are built into Twinkl Secondary Science teaching materials too. Our specially designed lessons focus on each of the reading AOs, helping you roll out these concepts to KS3 pupils (or re-enforce understanding at KS4).

AQA-Style Biology Paper 1 Walking Talking Mock Exam Pack

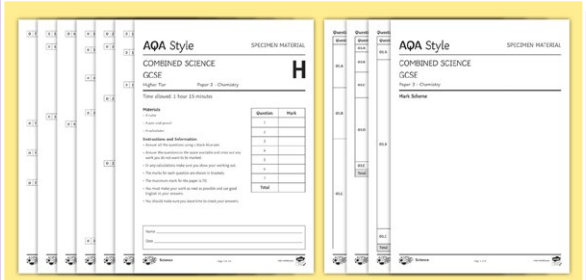




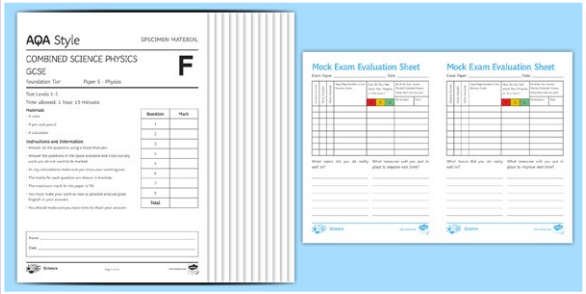
AQA Style Biology Specimen Paper: Paper 1 Foundation



AQA Style Chemistry Specimen Paper: Paper 3 Higher



AQA Style Physics Specimen Paper: Paper 5



Lesson Activities

Innovative and exciting science resources

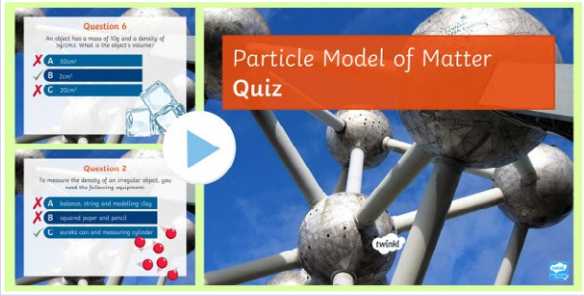
Looking for fresh ideas to build into your lessons? Or new ways to enhance learning? Then take a look at our stand-alone lesson activities.

Some of our most popular examples include our KS3 revision packs, quick quizzes, match and draw and find a friend. Remember, these fantastic resources are all created by Twinkl's team of experienced secondary teachers, so you'll always find quality classroom-ready content you can trust!

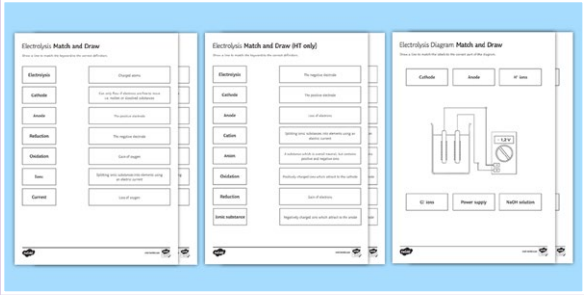
KS3 The Earth Revision Pack



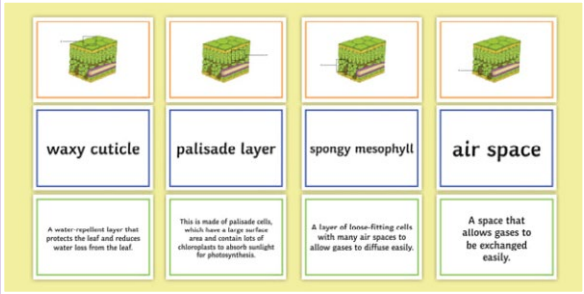
Particle Model of Matter Quick Quiz



Electrolysis Differentiated Match and Draw



Leaf Structure and Function 'Find a Friend' Card Sort



Teacher Toolbox

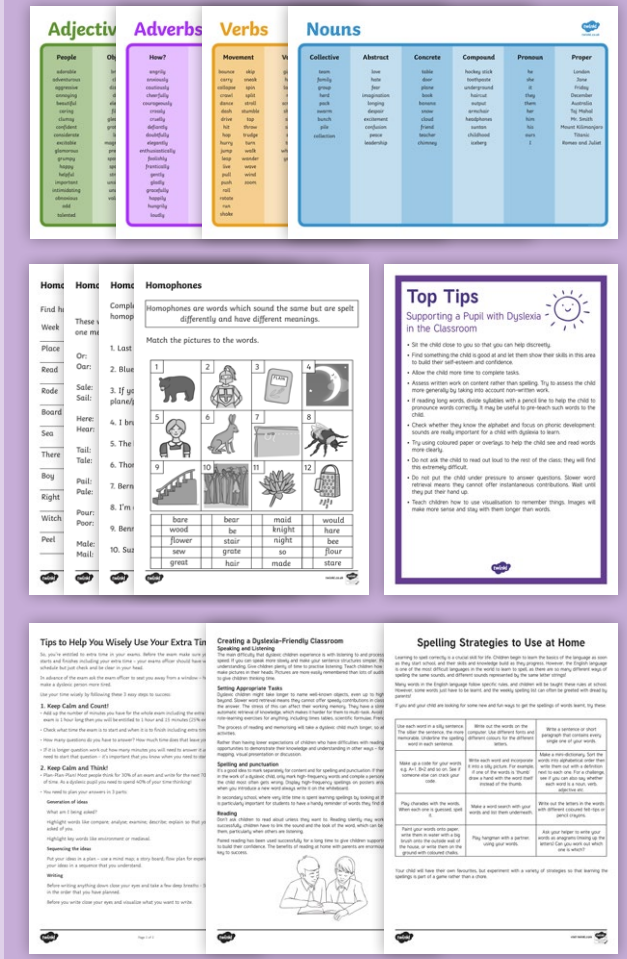
Helping you develop as a teacher

As well as the amazing Twinkl Secondary Science classroom resources, you can also access content designed to help enhance your own teaching and pedagogy. This exclusive Teacher Toolbox gives you hundreds of items designed to help your teaching practice. This includes:

- guidance on working with SEN pupils;
- marking tools;
- access arrangements;
- whole school literacy documents, plus more.

We're here to support your teaching practice too, which is why your Twinkl Secondary Science subscription includes resources that help you prioritise your professional development.

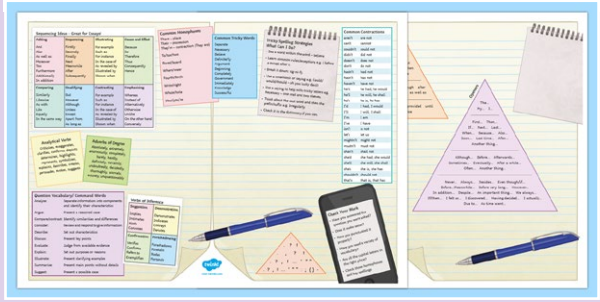
Supporting Dyslexia in KS3 and KS4 Resource Pack



Case File Recording and Assessment Documents



Whole School Literacy Mat



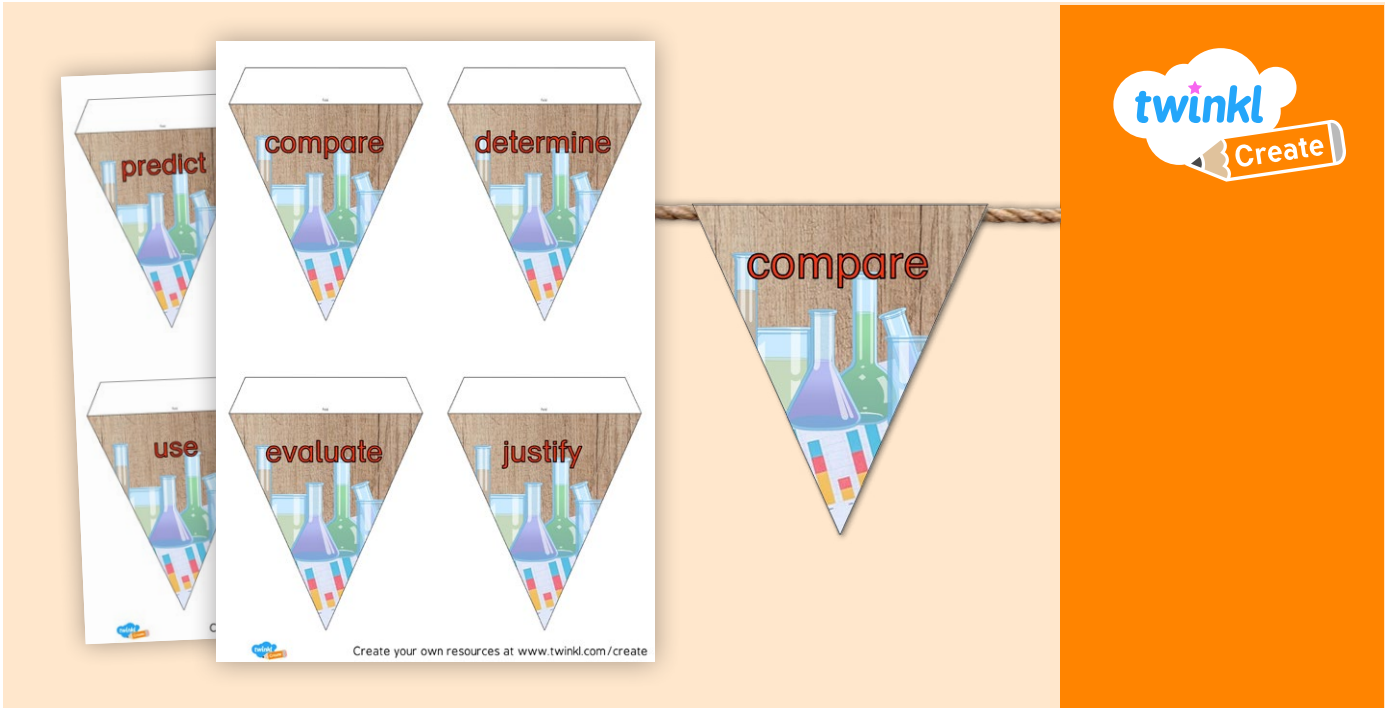
Gold Silver Bronze Certificates



Twinkl Create

Give your content the Twinkl touch

Twinkl Create is an innovative tool that lets you create your own resources online using Twinkl designs and artwork. You can drag images from any of our content onto a number of different templates to create new worksheets, banners, display material, and more.



Get creative! Blend your own lesson content with Twinkl's suite of visual materials to produce your own engaging resources.



Created by Teachers

Meet our experienced team



Becky

Becky has been a science teacher for over 10 years, becoming a head of biology and KS3. She loves spending time making fun and engaging resources that allow pupils to learn through practical science.



Claire

Claire has been a teacher for 9 years and has taught in a variety of roles. She's always on the lookout for ways to contextualise topics to make them fun, relevant and engaging.



Amanda

While completing her PhD in genetics, Amanda supported biology undergraduates and contributed to the university outreach programme for sixth form students. She quickly developed a passion for teaching. Having taught secondary school science for a number of years, Amanda now works as a freelance tutor.



Fiona

Fiona has been teaching in various roles for 9 years. She loves finding innovative and exciting ways to engage and inspire students in science, helping them to see the real-world applications of their learning.



Sue

Sue has been teaching secondary science for over twenty-five years and has recently started to coordinate the science curriculum at a local primary school. Delivering science in an exciting, challenging and innovative manner has always been a priority.

Join the Secondary Community

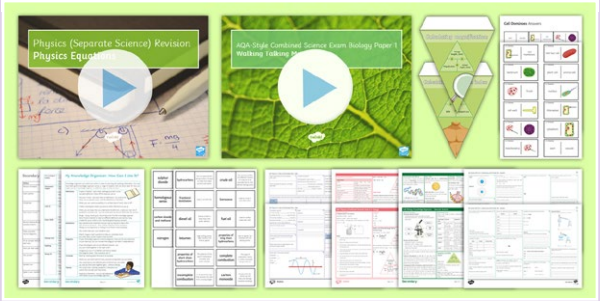
Access Secondary Science today!

Whet your appetite with FREE Secondary Science taster packs.

Free Secondary Science Taster Resource Pack



KS4 Science Revision Taster Resource Pack



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Secondary solutions at your fingertips

Secondary Science

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